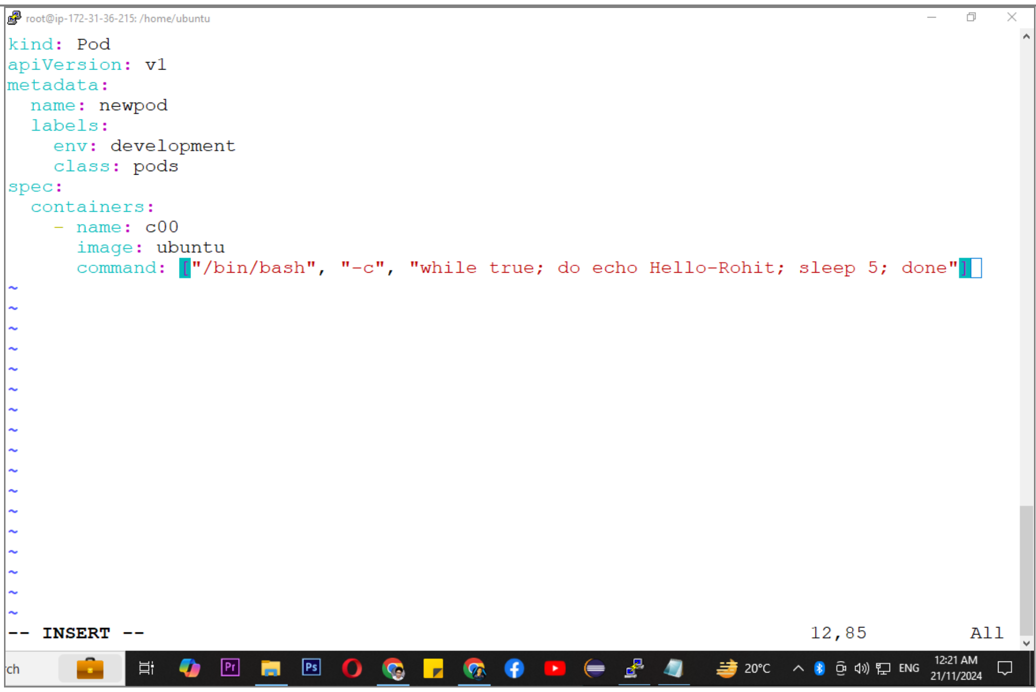
Labels and Selectors in Kubernetes

Rohit Surwade

Manifest with labels

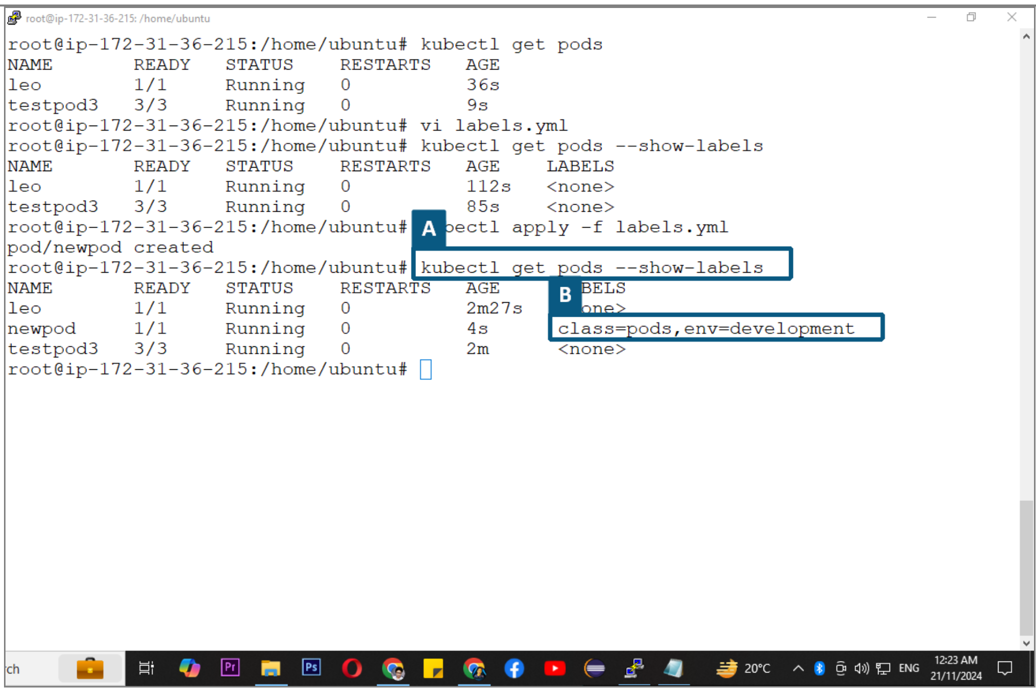


1.

(This manifest defines a Kubernetes Pod named newpod that runs a single container based on the ubuntu image.
The container executes a command that continuously prints "Hello-Rohit" every 5 seconds.
Labels are key-value pairs that can be used to organize and select resources. In this case, the Pod has two labels: env: development and class: pods.)

kind: Pod
apiVersion: v1
metadata:
name: newpod
labels:
env: development
class: pods
spec:
containers:
- name: c00
image: ubuntu
command: ["/bin/bash", "-c", "while true; do echo Hello-Rohit; sleep 5; done"]

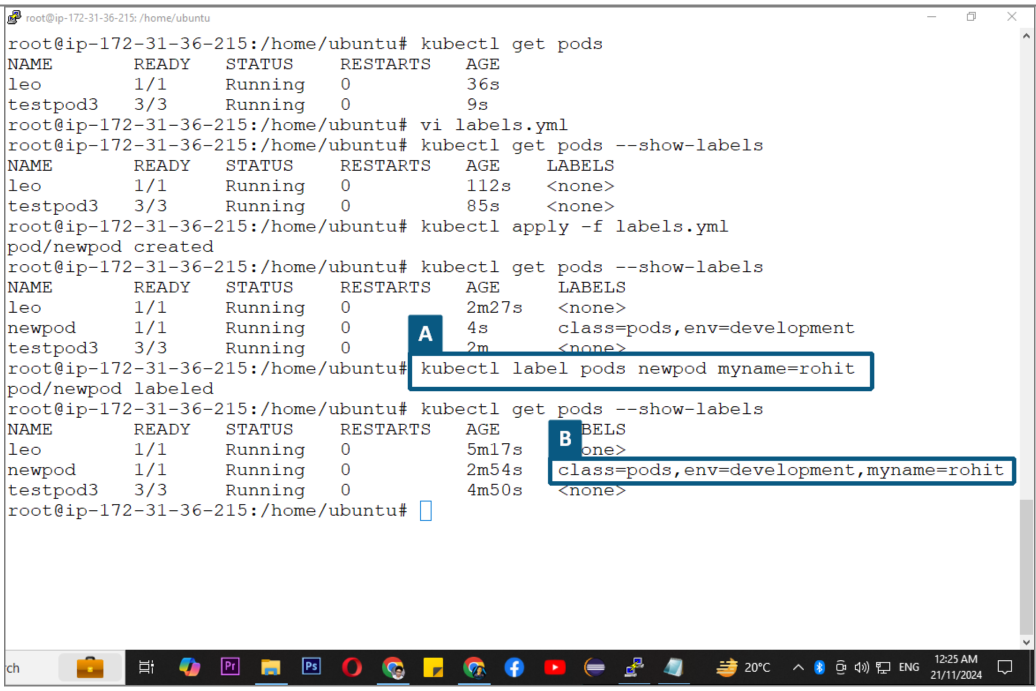
How to check the lables



2.

kubectl get pods --show-labels

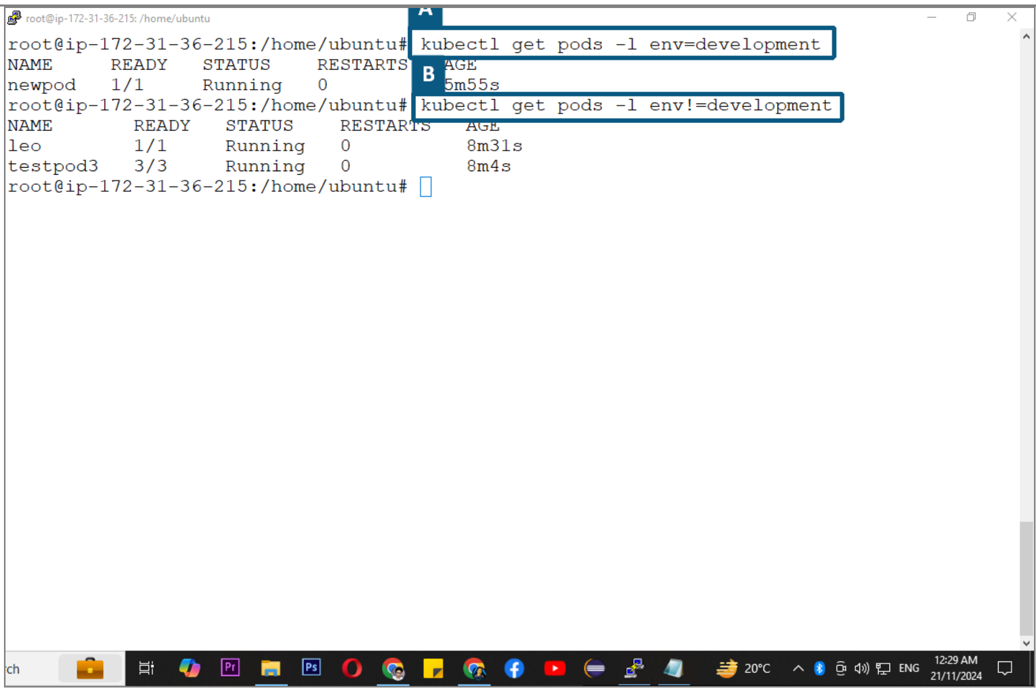
How to give labels using Imperative method



3.

kubectl label pods pod_name key=value (for ex: myname=rohit)

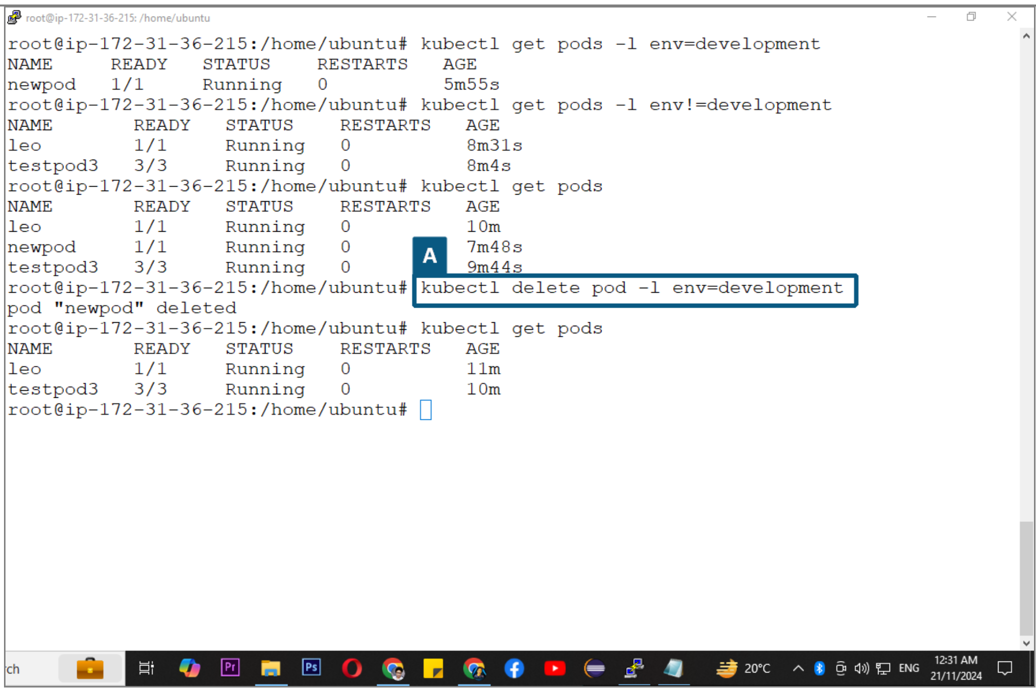
Equality based Label Selectors(=, !=)



4.

kubectl get pods -l key=value
kubectl get pods -l key!=value

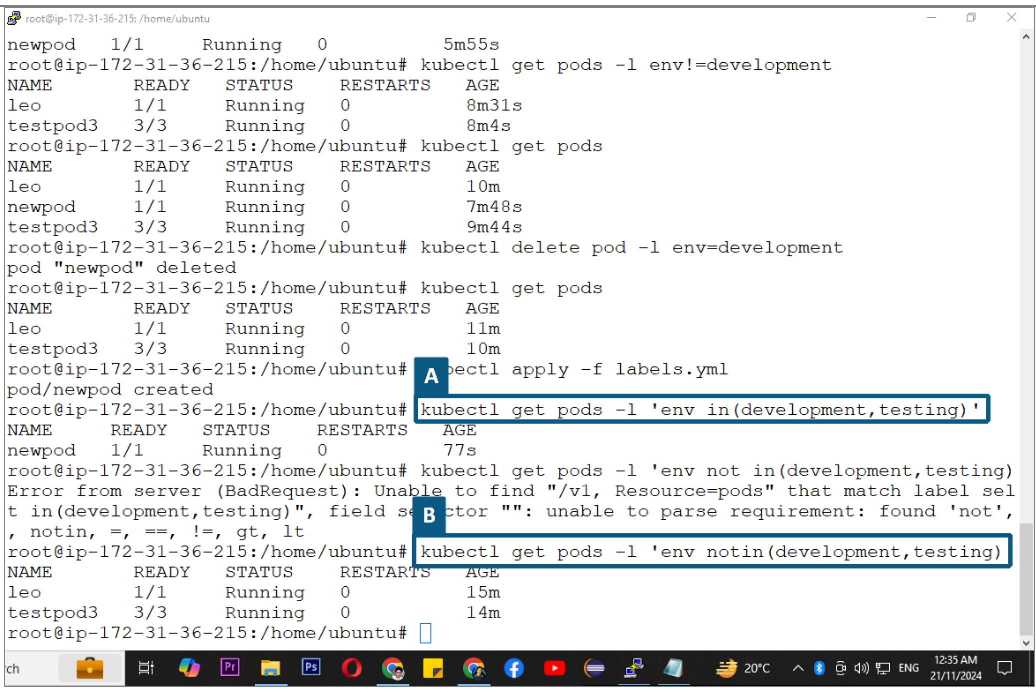
How to delete pods using lables & selectors



5.

kubectl delete pod -l key=value

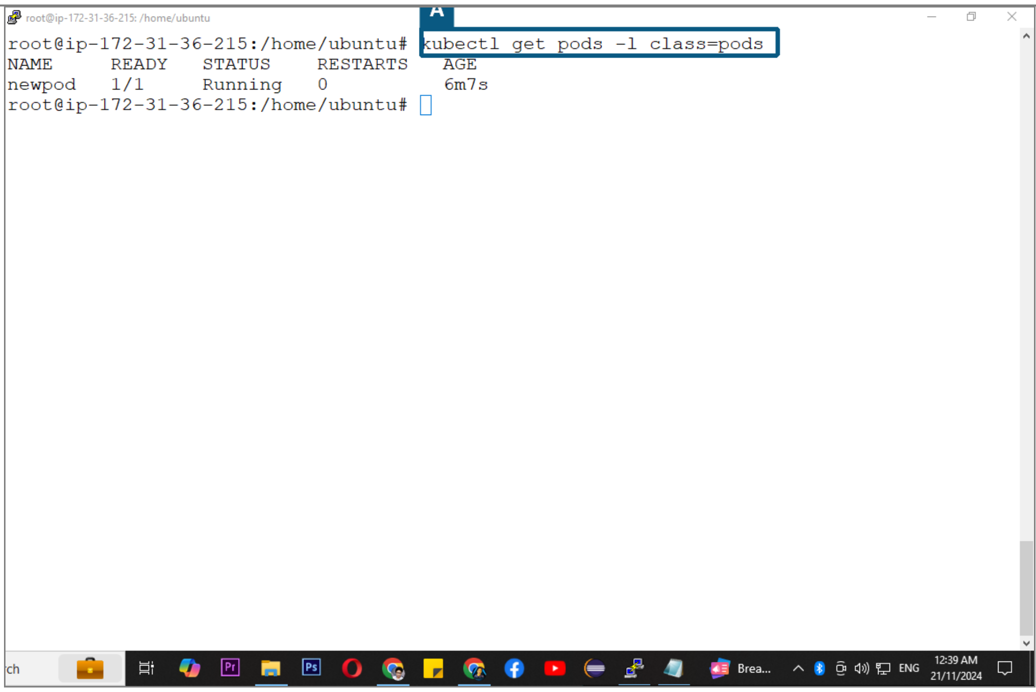
Set based Label Selectors(in, notin, exists, doesnotexist)



6.

kubectl get pods -l 'key in(value1,value2)'
kubectl get pods -l 'key notin(value1,value2)'

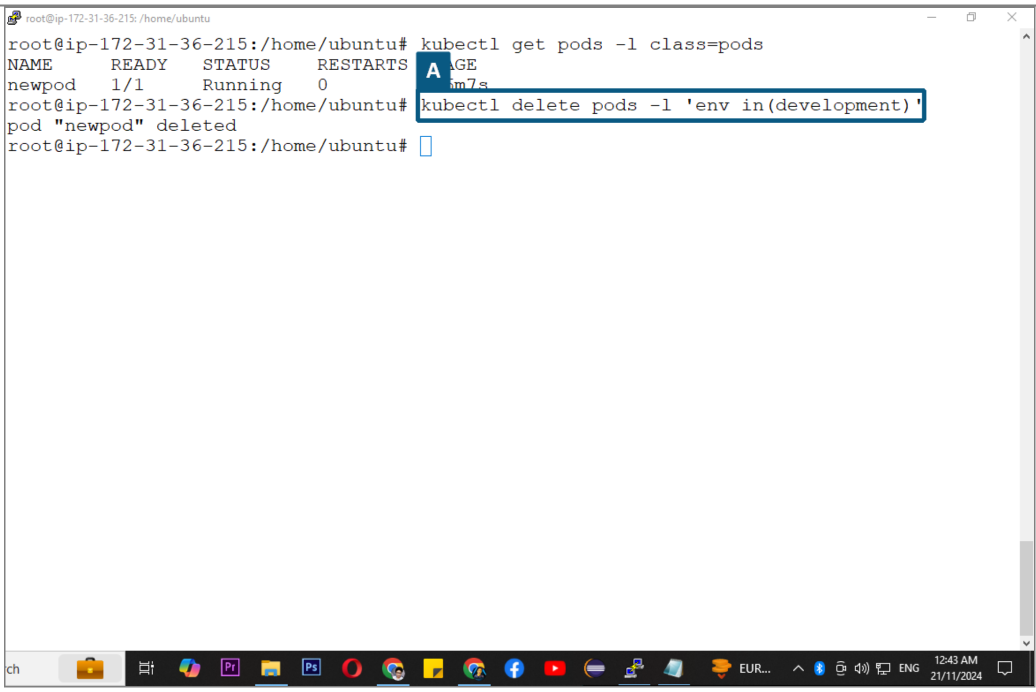
How to search for pods using labels and selectors



7.

kubectl get pods -l class=pods

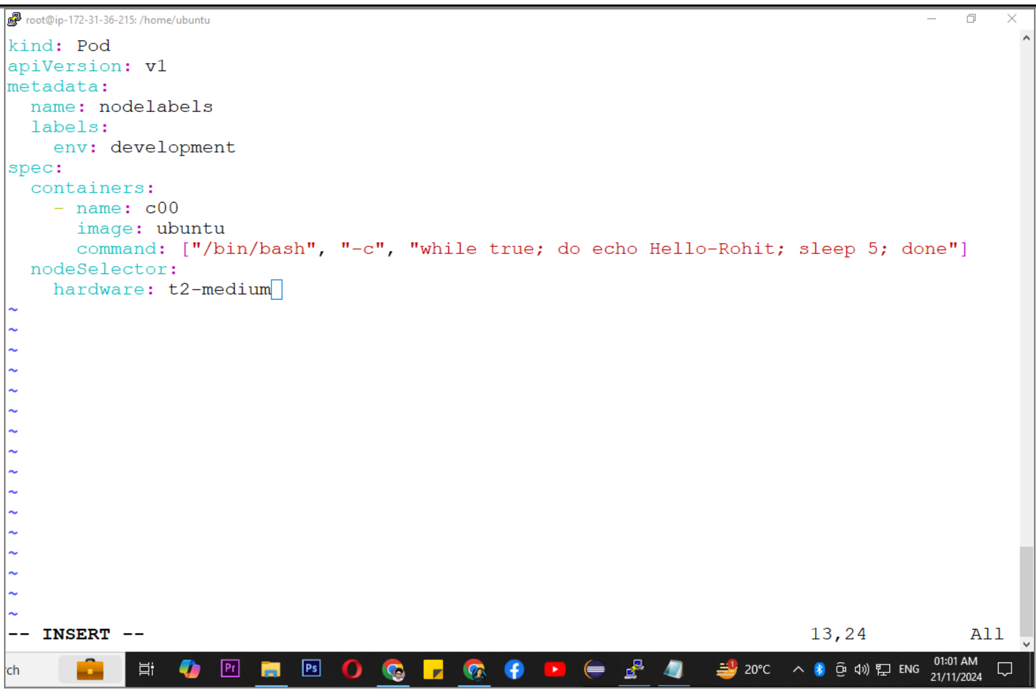
How to delete pods using set based label selectors



8.

kubectl delete pods -l 'key in(value)'

Manifest with a nodeSelector

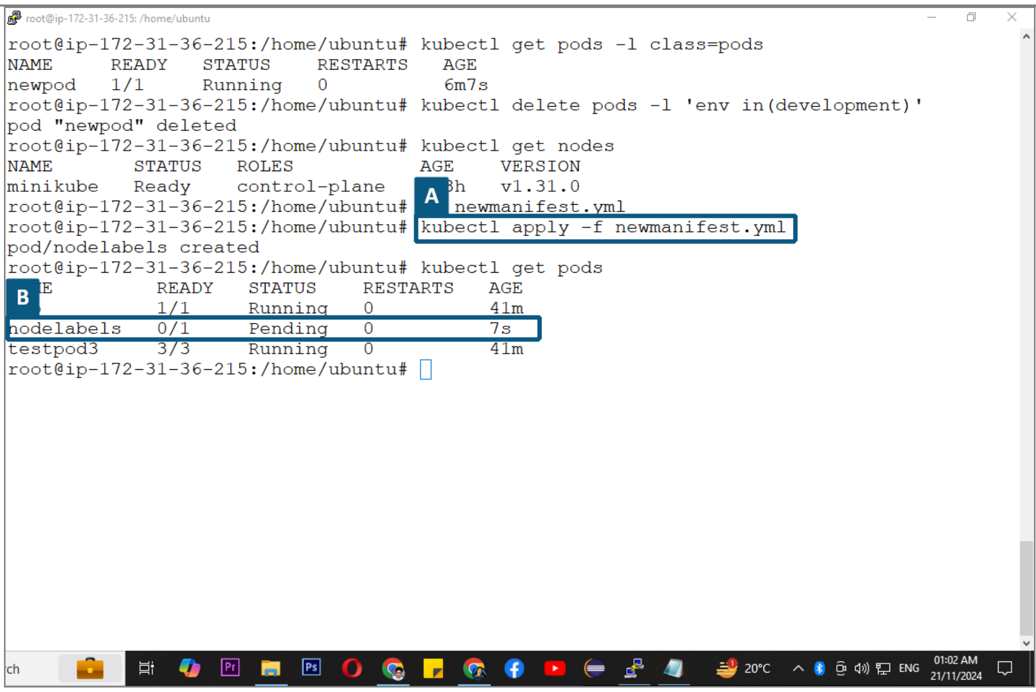


9.

(This manifest defines a Kubernetes Pod named nodelabels that runs a single container based on the ubuntu image.
The container executes a command that continuously prints "Hello-Rohit" every 5 seconds.
The Pod is scheduled on nodes labeled with hardware: t2-medium.This specifies the node selection criteria for the Pod.
The Pod will be scheduled on nodes that have the label hardware: t2-medium.
Make sure that you have nodes in your cluster with this label.)

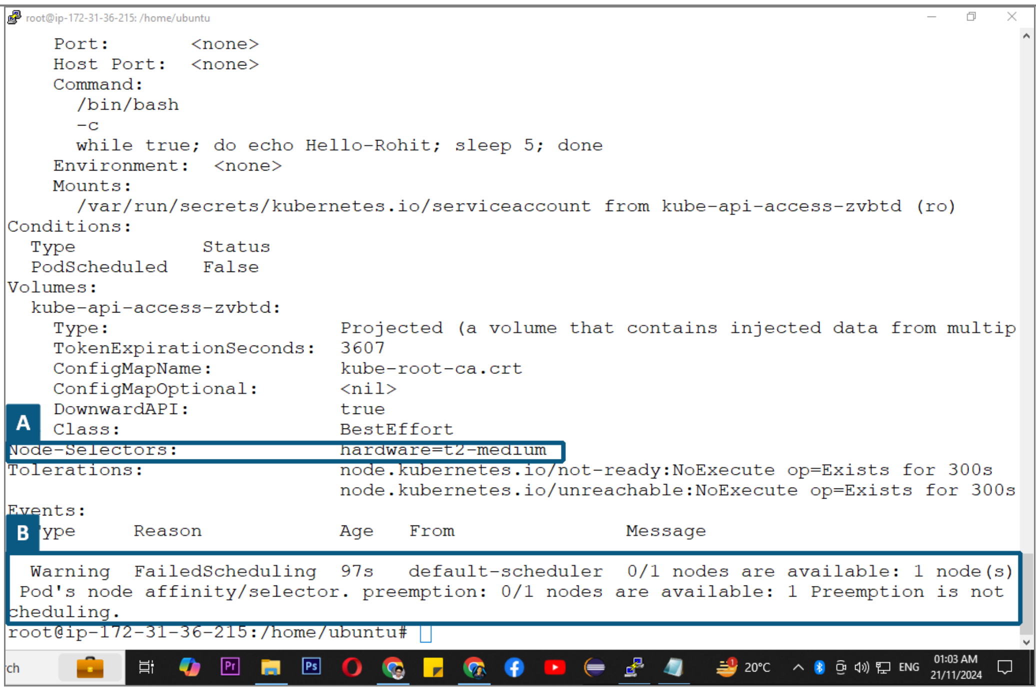
kind: Pod
apiVersion: v1
metadata:
name: nodelabels
labels:
env: development
spec:
containers:
- name: c00
image: ubuntu
command: ["/bin/bash", "-c", "while true; do echo Hello-Rohit; sleep 5; done"]
nodeSelector:
hardware: t2-medium

Error creating the pod (nodeSelector)



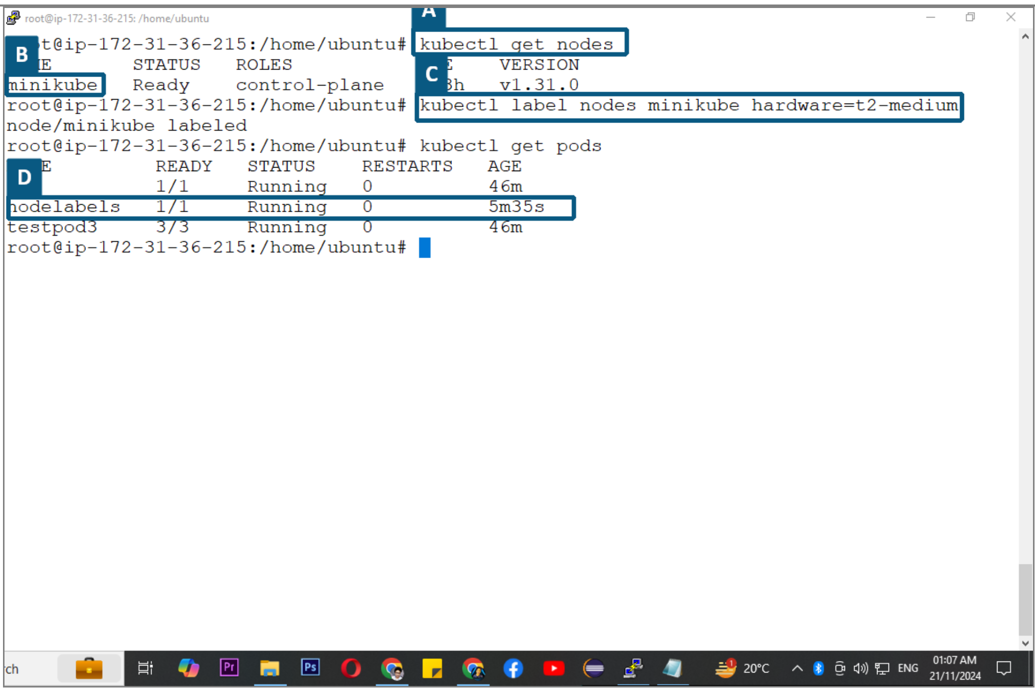
10.

Node cannot be scheduled cause we haven't added the nodeSelector hardware label to the node



11.

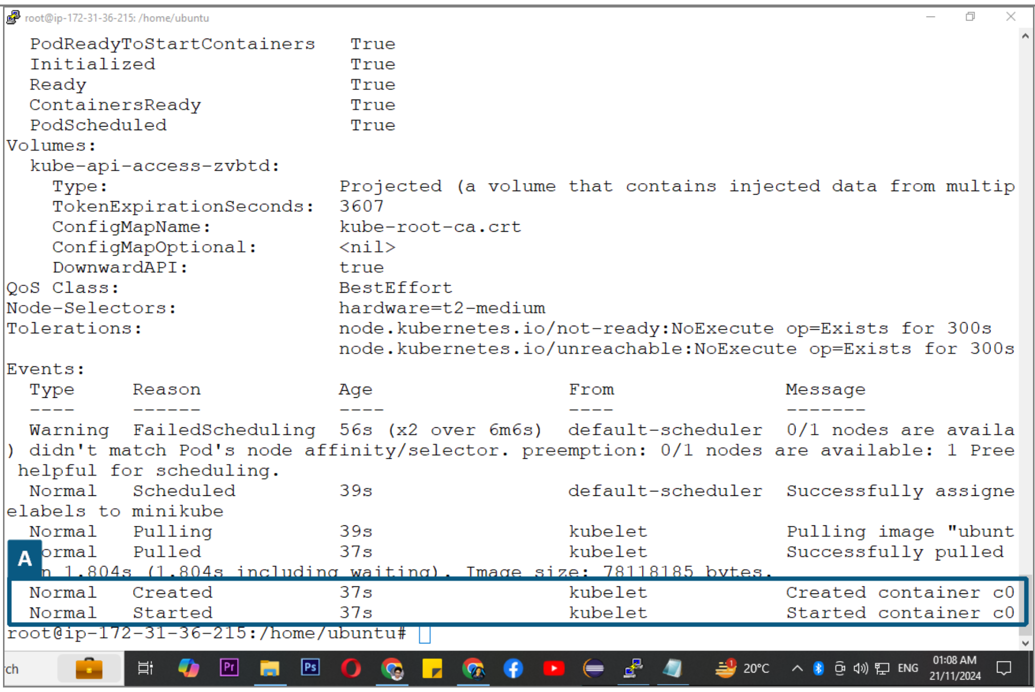
How to add the nodeSelector hardware label to the node



12.

kubectl get nodes
kubectl label nodes node_name hardware=t2-medium(key-value)

Node is successfully scheduled and container is created



13.